MARKING SCHEME

Model Paper — 06

Mathematics I

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Note (for teachers):

- Part A: 2 marks per question (1 mark for the correct method, 1 mark for the correct answer). Total 50 marks.
- Part B: 10 marks per question. Total 50 marks. The marks for each step are given in the right-hand column.
- Give full marks for correct alternative methods. Accept reasonable readings from graphs and constructions (e.g. ± half a small square).

Part A — Answers

1. Work = 12 × 20 = 240 worker-days; workers needed for 15 days = 240 ÷ 15 = 16; extra workers = 16 − 12

Final answer: 4 workers

(2 marks)

2. Draw a line through E parallel to AB (and CD): $A\hat{E}C = B\hat{A}E + E\hat{C}D$ (alternate angles); $B\hat{A}E = 95° - 40°$

Final answer: 55°

(2 marks)

3. LCM = 12x: $\frac{10 + 9 - 6}{12x} = \frac{13}{12x}$

Final answer: $\frac{13}{12x}$

(2 marks)

4. The faces of a right triangular prism are triangles and rectangles (a rectangle may be a square). A parallelogram that is not a rectangle cannot be a face.

Final answer: (iii)

(2 marks)

5. 20° ↔ 30 packets, so 1° ↔ 1.5 packets; angle for dhal = 360° − 180° − 20° = 160°; packets of dhal = 160 × 1.5

Final answer: 240 packets

(2 marks)

6. ADBE is cyclic: $A\hat{D}B = 180° - 150° = 30°$. C is the midpoint of AB and DOC passes through O, so DC ⊥ AB and DC bisects $A\hat{D}B$: $C\hat{D}B = 30° \div 2$

Final answer: 15°

(2 marks)

7. $7(x^2 - 4) = 7(x^2 - 2^2)$

Final answer: 7(x + 2)(x − 2)

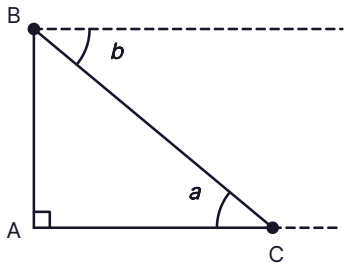
(2 marks)

8. $5400 = 5.4 \times 10^3$; $\lg 5400 = 3 + \lg 5.4 = 3 + 0.7324$

Final answer: 3.7324

(2 marks)

9. The horizontal through B is parallel to AC, so the alternate angles are equal. a = angle ACB (between CA and CB); b = angle between the horizontal through B and BC (see figure).



Final answer: $a = b$ (a and b correctly marked on the figure)

(2 marks)

10. Curved surface area = $2\pi r \times 7r = 14\pi r^2$; base area = πr^2 ; ratio = $14\pi r^2 : \pi r^2$

Final answer: 14 : 1

(2 marks)

11. Total outcomes = $4 \times 4 = 16$; sum 8: (2, 6), (6, 2), (3, 5), (5, 3) → 4 outcomes; P = 4/16

Final answer: $\frac{1}{4}$

(2 marks)

12. Let $B\hat{A}D = p$. AB = BD gives $B\hat{D}A = p$, so the exterior angle $D\hat{B}C = 2p$; BD = DC gives $D\hat{C}B = 2p$. In triangle ADC: $p + 2p + 105° = 180° \Rightarrow p = 25°$; $x = 180° - 2(50°)$

Final answer: x = 80°

(2 marks)

13. $A \cap B = \{3, 6\}$, $A \cup B = \{1, 2, 3, 4, 6, 9\}$, $n(B') = 10 - 5 = 5$; only (ii) is correct.

Final answer: (ii)

(2 marks)

14. $(24a^4b^3) \div (8a^2b)$

Final answer: $3a^2b^2$

(2 marks)

15. $r = (-12) \div (-24) = \frac{1}{2}$; $T_4 = -6$; $T_5 = -3$

Final answer: −3

(2 marks)

16. $B\hat{D}C = B\hat{A}C = 35°$ (angles in the same segment); $A\hat{D}C = 2 \times 35° = 70°$; $C\hat{B}E = A\hat{D}C$ (exterior angle of a cyclic quadrilateral)

Final answer: 70°

(2 marks)

17. HCF of the coefficients = 3; lowest powers p, q^2

Final answer: $3pq^2$

(2 marks)

18. $\pi r^2 h = 1386 \times h \Rightarrow r^2 = 1386 \times \frac{7}{22} = 441$

Final answer: r = 21 cm

(2 marks)

19. Midpoint theorem: DE ∥ BC, BC = 2 × DE = 30 cm; AB = 16 cm so DB = 8 cm; $AE^2 = 8^2 + 15^2 \Rightarrow$ AE = 17 cm so EC = 17 cm; perimeter = 8 + 30 + 17 + 15

Final answer: 70 cm

(2 marks)

20. Gradient = $\frac{10-4}{6-2} = \frac{3}{2}$; $y - 4 = \frac{3}{2}(x - 2)$

Final answer: $y = \frac{3}{2}x + 1$ (3x − 2y + 2 = 0)

(2 marks)

21. AB is a diameter, so $A\hat{D}B = 90°$; $D\hat{A}B = B\hat{C}D = 56°$ (opposite angles of a parallelogram); $A\hat{B}D = 180° - 90° - 56° = 34°$; $D\hat{B}E = 180° - 34°$

Final answer: 146°

(2 marks)

22. Rate 8%: interest of year 2 = 8% of the amount after year 1, so amount after year 1 = 432 × 100 ÷ 8 = Rs 5400; P × 1.08 = 5400

Final answer: Rs 5000 (check: $5000 \times 1.08^2 = 5832$)

(2 marks)

23. CB = CD (radii): $C\hat{B}D = (180° - 80°) \div 2 = 50°$; AC ∥ BD: $A\hat{C}B = C\hat{B}D = 50°$ (alternate angles); $A\hat{E}B = ½ \times A\hat{C}B$

Final answer: 25°

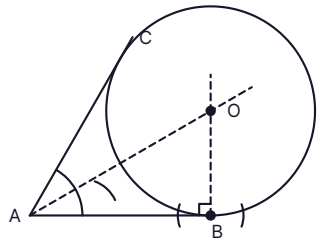
(2 marks)

24. $2x > 1 - 9 \Rightarrow 2x > -8 \Rightarrow x > -4$

Final answer: −3, −2, −1

(2 marks)

25. Complete the bisector of BÂC: arc from the second marked point meeting the first arc, join A to the intersection (this gives BÂO = 30°, OA bisects BÂC). At B draw arcs of equal radius from the two marked points on AB, meeting above AB, and join to B: the perpendicular to AB at B. O is where the bisector meets the perpendicular (OB ⊥ AB). Marks: constructions completed 1; O marked at the intersection 1.



Final answer: O = intersection of the angle bisector at A and the perpendicular at B

(2 marks)

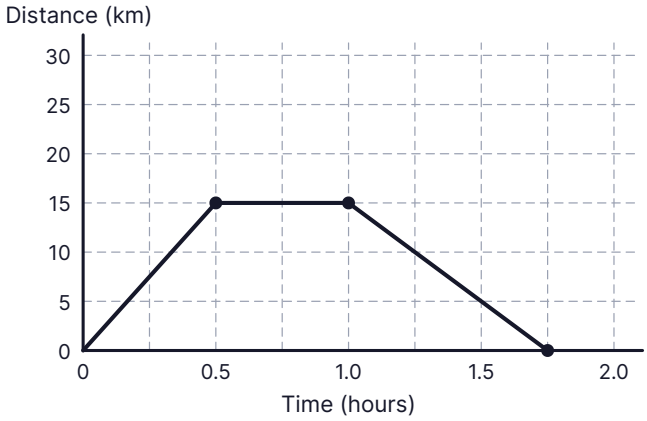
Part B — Answers and mark allocation

Question 1

(i)	
$1 - \frac{1}{6} = \frac{5}{6}$	**1**
(ii)	
Other expenses = $\frac{3}{4} \times \frac{5}{6} = \frac{5}{8}$	**1**
Remaining = $\frac{5}{6} - \frac{5}{8} = \frac{5}{24}$	**1**
(iii)	
$\frac{5}{24}$ of the money received = Rs 40 000 ⇒ money received = 40 000 × $\frac{24}{5}$ = Rs 192 000	**1**
Food = $\frac{1}{6}$ × 192 000 = Rs 32 000 ; other expenses = $\frac{5}{8}$ × 192 000 = Rs 120 000	**1**
(iv)	
Tax paid = S − 192 000 (S = salary)	**1**
S − 192 000 = $\frac{10}{100}$ (S − 120 000)	**1**
100S − 19 200 000 = 10S − 1 200 000 ⇒ 90S = 18 000 000 ⇒ S = Rs 200 000	**1**
(v)	
$\frac{32\,000}{200\,000} \times 100$	**1**
= 16 %	**1**

Total marks: 10

Question 2



(a)(i)	
Horizontal line from (0.5 h, 15 km) to (1.0 h, 15 km) (staying 30 minutes)	**1**
Straight line from (1.0 h, 15 km) to (1.75 h, 0 km) (return in 45 min); see figure	**1**
(a)(ii)	
Speed = 15 km ÷ $\frac{3}{4}$ h	**1**
= 20 km h^{-1}	**1**
(b)(i)	
Perimeter = arc AEB + AD + DC + CB (AB is not part of the boundary): AD = CB = 11 cm, DC = 14 cm	**1**
Arc AEB = $\frac{1}{2} \times 2 \times \frac{22}{7} \times 7$ = 22 cm	**1**
Perimeter = 22 + 11 + 14 + 11 = 58 cm	**1**
(b)(ii)	
Area of semicircle = $\frac{1}{2} \times \frac{22}{7} \times 7 \times 7$ = 77 cm^2	**1**
Area of rectangle = 14 × 11 = 154 cm^2	**1**
154 ÷ 77 = 2 times	**1**

Total marks: 10

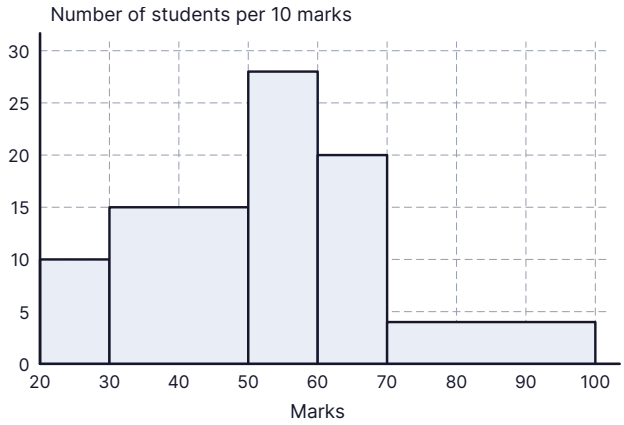
Question 3

(i)	
Rates = $\frac{14}{100} \times 120\,000$	**1**
= Rs 16 800	**1**
(ii)	
Annual rent = 15 000 × 12 = Rs 180 000	**1**
Maintenance = $\frac{10}{100} \times 180\,000$ = Rs 18 000	**1**
Left = 180 000 − 16 800 − 18 000 = Rs 145 200	**1**
(iii)(a)	
Number of shares = 12 100 ÷ 5	**1**
= 2420	**1**
(iii)(b)	
Total invested = 2420 × 100 = Rs 242 000	**1**
Additional money = 242 000 − 145 200 = Rs 96 800	**1**
$\frac{96\,800}{242\,000} \times 100 = 40\ \%$	**1**

Total marks: 10

Question 4

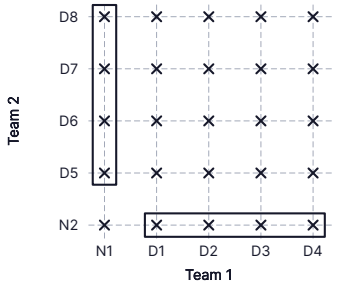
Marks	Number of students	Students per 10 marks	Cumulative frequency
20 – 30	10	10	10
30 – 50	30	15	40
50 – 60	28	28	68
60 – 70	20	20	88
70 – 100	12	4	100

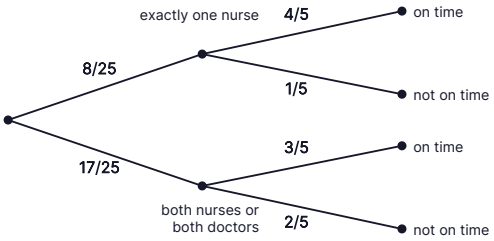


(i)	
Reading the polygon: 30 (class 30 – 50)	**1**
28 (class 50 – 60)	**1**
20 (class 60 – 70) and 12 (class 70 – 100)	**1**
(ii)	
Heights = frequency ÷ class width × 10: $\frac{30}{20} \times 10 = 15$, 28, 20, $\frac{12}{30} \times 10 = 4$	**1**
Bars drawn on the correct class boundaries (30 – 50, 50 – 60, 60 – 70, 70 – 100), touching each other	**1**
Bar heights 15, 28, 20, 4 drawn correctly (± half a small square)	**1**
(iii)	
Cumulative frequency: 40, 68, 88, 100 (all four correct: 2 marks; three correct: 1 mark)	**2**
(iv)	
Students with 50 or more marks = 28 + 20 + 12 = 60	**1**
Probability = $\frac{60}{100} = \frac{3}{5}$	**1**

Total marks: 10

Question 5





(i)	
All 25 outcomes marked with × at the intersections (24 or 23 correct: 1 mark)	**2**
(ii)	
Closed curves round the 4 outcomes (N1 with D5, D6, D7, D8) and the 4 outcomes (D1, D2, D3, D4 with N2): 8 outcomes	**1**
n(E) = 4 + 4 = 8	**1**
P(exactly one nurse) = $\frac{8}{25}$	**1**
(iii)	
Both nurses (1 × 1 = 1) or both doctors (4 × 4 = 16): $\frac{17}{25}$; first-stage branches $\frac{8}{25}$, $\frac{17}{25}$	**1**
Complements $\frac{1}{5}$ and $\frac{2}{5}$ for “not on time”	**1**
Given values $\frac{4}{5}$ and $\frac{3}{5}$ written on the correct branches (see tree)	**1**
(iv)	
P(not on time) = $\frac{8}{25} \times \frac{1}{5} + \frac{17}{25} \times \frac{2}{5}$	**1**
$\frac{8}{125} + \frac{34}{125} = \frac{42}{125}$	**1**

Total marks: 10